



Prescient AI

Background and instructions

Prescient AI provides marketing mix modeling results to omni-channel brands seeking to better understand their marketing attribution. It's paramount that the data we feed into our models is consistent and sensical. It's equally important to have modeled outputs that are reasonable and make sense in the context of the campaigns and brand.

This case study was designed to take around 3-4 hours to complete, but you are free to take the time you need. The goal of this assignment is to assess your analytical abilities, reasoning, creativity, and communication.

Overall deliverables:

- A code-based analysis to compare and contrast various datasets.
- Presenting your findings from the above to the Prescient team.
- 2-3 questions on marketing mix modeling (conceptual only) during the interview.

Here are a few specific things we will be looking for:

- Ability to summarize data accurately.
- Clear communication that non-technical stakeholders can understand.
- Code structure/style that teammates can navigate and understand.
- Critical thinking and creativity in problem solving.

Submitting your work

Please submit all code used to complete your analysis (e.g., Python files, Notebooks, etc). In addition, you will be presenting your findings to the team—so if you are using a presentation file format (Google Slides, etc), please also share the presentation file/link. Please keep your presentation short — it should take no more than 5-10 minutes to present your findings.

All resources can be shared with philip@prescient-ai.io

Metric Analysis & Quality Assurance

Imagine the data science team has been asked to compare modeling input and output between development and production environments. You have been tasked with preparing an analysis of the discrepancies between the development and production environments for company "X".

Schema/terminology information:

- **Reported Date:** *Date (YYYY-MM-DD) the data/record was created.*
- **Company:** *The name of the company.*
- **Marketing Channel:** *The platform of the advertisement/campaign (e.g., Facebook, Google, Pinterest, etc.)*
- **Campaign ID:** *The unique identifier for a given campaign.*
- **Campaign Name:** *The name of the campaign assigned by the client (company X).*
- **Spend:** *Daily amount marketers spent on a given campaign.*
- **Channel reported revenue:** *Daily revenue reported from a given marketing channel (e.g., Facebook Ads, Google Ads, etc.) for a given campaign. Note: this revenue is subject to over- or under-reporting by the channels.*
- **First order revenue:** *This is modeled revenue. It represents revenue that comes directly through paid ads.*
- **Second order revenue:** *This is also modeled revenue. It represents revenue from consumers directly visiting a website/online store because they were influenced by a broader marketing effort.*
- **Total modeled revenue:** *First order revenue + second order revenue.*
- **Modeled vs reported metrics:** *At Prescient, we classify metrics we receive directly from our customers/their marketing platforms as reported. Whereas modeled metrics are what we create at Prescient through modeling or aggregation.*

Data:

- company_x_dev_reported_metrics.csv
- company_x_dev_modeled_metrics.csv
- company_x_prod_reported_metrics.csv
- company_x_prod_modeled_metrics.csv